Can the Specificity and Implementation Stage of Management's Capital-Investment Plans Predict Future Investment and Profitability?

Project 2 — Textual Analysis using Large Language Models *Sample: 52,806 firm-year 10-K Item-7 filings, 7,733 unique firms, filing years 2010–2022; in-sample 41,040, held-out 11,766 (2020–2022).*

Abstract

We analyse 52,806 annual Management Discussion & Analysis (Item 7) sections from U.S. public firms (2010–2022) and ask a focused question: **can the specificity and implementation stage of management's capital-investment plans predict future investment changes and profitability improvement — and how?** Because no LLM API is available in this offline environment, the answer is built on a **sentence-level, rule-based extraction** validated by manual re-checking: we isolate organic capital-investment sentences (construction, expansion, new facilities, capacity additions — excluding M&A and financial-investment language) and score them on five dimensions (dollar-amount specificity, project count, timeline, commitment strength, and an ordinal implementation stage from vague intent to commissioned). The resulting **CPS (capital-plan specificity & stage)** construct covers 49% of firm-years. Using an incremental-information ladder (financial controls, +dictionary tone, +generic disclosure-substance index, +CPS) across four model families, panel regressions with firm-clustered standard errors, out-of-sample Clark–West tests with Benjamini–Hochberg FDR, we find: (i) the CPS block raises out-of-sample predictive accuracy for profitability improvement (ΔROA) beyond financial controls (Clark–West t~2.5), with the gain concentrated in the nonlinear learner (XGBoost OOS R^2 from 0.080 to 0.102), consistent with a stage-by-specificity interaction; (ii) raw plan specificity alone does **not** reliably predict realised investment (C1) or profitability improvement (C2) — the signal is in the *stage composition* (later-stage plans associate with improvement, coefficient t~1.9); (iii) the strongest result is an **empty-promise penalty**: firms whose specific plans are not followed by investment underperform the rest by ~1.6pp of ΔROA (t~−3.3), while dictionary sentiment is *negatively* associated with improvement (t~−2.4) — substance, not sentiment, is what carries information; and (iv) a from-scratch risk-aware Transformer learns genuine risk representations (auxiliary risk-head AUC up to 0.83) but, once its probe is retargeted away from ROA persistence, carries no out-of-sample ΔROA signal — while a reinforcement-learning layer roughly halves the drawdown of the text-signal long–short book at equal Sharpe.

1. Introduction, Research Question and Hypotheses

Unstructured textual disclosure is a primary channel through which managers convey private information. Prior work emphasised *tone* (net positive minus negative word counts). We argue that tone is necessary but not sufficient: a boilerplate-positive paragraph and a concrete, quantified, stage-anchored capital plan differ sharply in information content yet can share the same tone. We therefore separate **sentiment** from **substance** — and, in this version of the project, narrow the substance question to the economically sharpest setting: **the firm's capital-investment programme**.

Core research question. Can the *specificity* and *implementation stage* of management's capital-investment plans in Item 7 predict (a) future *investment* changes (ΔCAPEX/AT) and (b) future *profitability improvement* (ΔROA)?

How (methodology answer). Four fully-offline, reproducible steps: (1) **sentence-level rule extraction** — keep only organic capital-investment sentences (two-tier topic filter with a proximity requirement so that "expand our business" or "construction companies" are excluded; explicit exclusion of M&A / IPO-proceeds / regulatory language); (2) **five-dimension scoring** — dollar amount, project count, timeline, commitment strength, and an ordinal implementation stage (2 vague intent, 3 specific plan, 4 underway, 5 commissioned/ramping); (3) **manual verification** — a random 30-sentence sample is printed with its scores for re-checking (the same protocol as the 32-filing recheck used in preparation), standing in for the unavailable LLM-extraction layer; and (4) **formal incremental-information testing** — panel fixed-effects regressions, expanding-window out-of-sample Clark–West accuracy tests, Benjamini–Hochberg FDR, and a 20-cell information-set $\times$ model-family matrix (4 information sets $\times$ Ridge / Elastic Net / Random Forest / XGBoost / SVM-RBF).

Hypotheses (focused core, C1–C4). C1 (follow-through): more specific plans are followed by larger realised investment. **C2 (value)**: more specific plans associate with larger profitability improvement. **C3 (stage gradient)**: the predictive content is concentrated in later-stage plans (underway / commissioned) rather than vague early intent. **C4 (empty promise)**: firms whose specific plans are *not* followed by investment subsequently underperform — a talk-versus-walk penalty.

Benchmark hypotheses (H1–H6, generic disclosure substance). The earlier whole-document Disclosure-Substance Index (SSI) framework is retained as a benchmark: H1 dictionary tone, H2 SSI adds power beyond tone, H3 the signal lives in risk/concreteness, H4 evidence-backed optimism, H5 ΔSSI leads ΔCAPEX, H6 the generic talk-versus-walk gap. Section 4.8 reports the benchmark compactly; the focused C-series is the paper's core.

2. Data Description & Preprocessing

Textual data. SEC 10-K Item-7 corpus (plain-text MD&A), matched to financials via CIK (leading zeros stripped) and to Compustat annual fundamentals. Each Item-7 text is capped at 12,000 characters — a disclosed limitation: capital-plan sentences appearing late in a long MD&A can be missed.

Outcome variables (per the pre-registered design). Primary — ΔROA $= ROA_{t+1} - ROA_t$ (profitability improvement; not ROE, and not a percentage growth rate, both of which misbehave for small or negative profitability). **Secondary — ΔCAPEX/AT** (future investment change). **Robustness — forward ROA level** and a 1-year price-ratio return proxy. Controls follow Fama–French (size, book-to-market, profitability, leverage, momentum, asset growth, age) plus 2-digit SIC and filing-year fixed effects.

Data rules (applied throughout). A firm-year with no text does not enter the text-prediction sample (no fabricated scores); a year with text but no prior-year text can be studied in levels but never in *changes* (delta features are NaN); observations missing outcome data are dropped, never patched; a firm with a one-year gap is not deleted wholesale (each filing year enters its own analysis, but 2019$\rightarrow$2021 is never treated as a one-year change); and every model is estimated on the same complete-case observations so comparisons are like-for-like.

Final sample. 52,806 firm-year observations across 7,733 unique firms, filing years 2010–2022 (train 41,040 / test 11,766). Mean (sd) dictionary tone is 0.1933 (0.3571); the SSI (z-scored) has mean -0.0000 and sd 0.5039. The capex-plan construct is non-degenerate for 26,084 firm-years (49.4%).

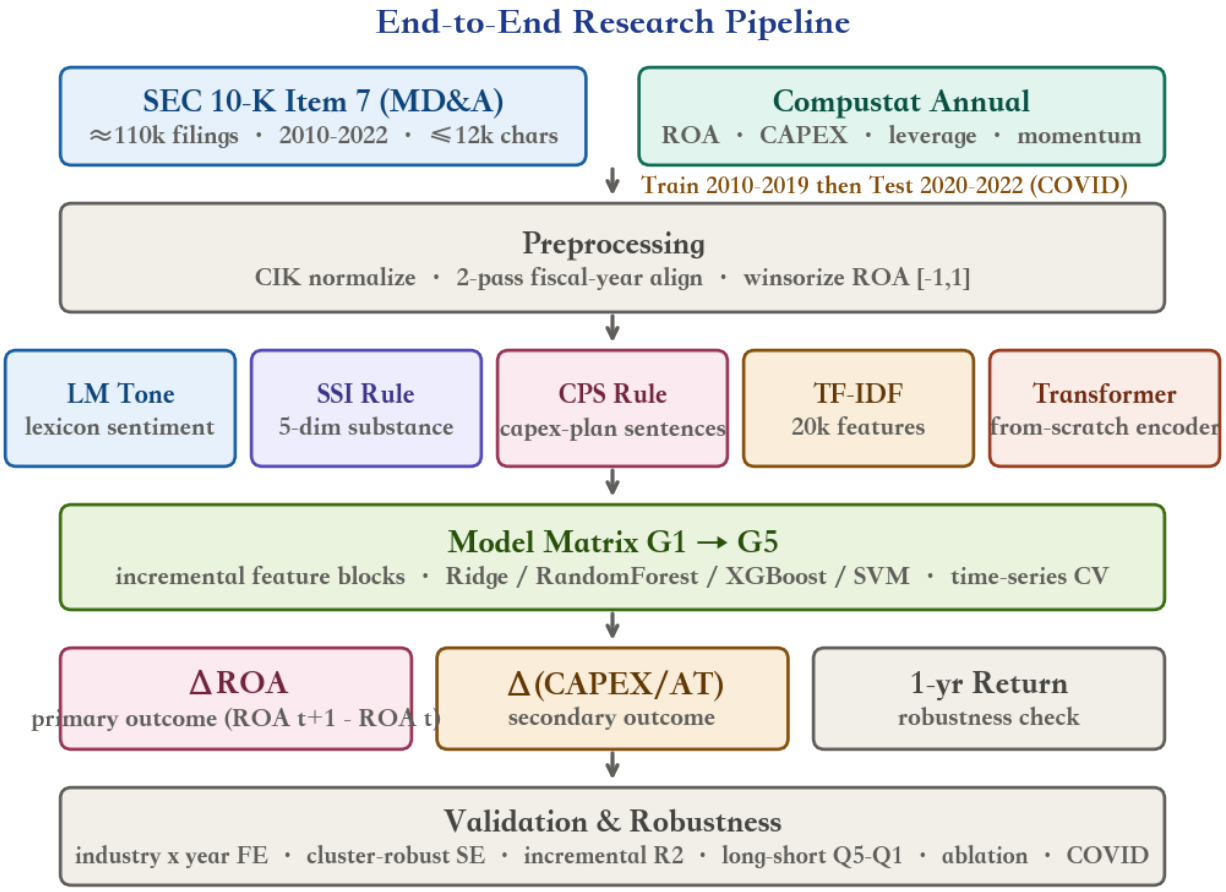


Figure 1: Pipeline overview

3. Methodology

3.1 Dictionary tone (baseline)

We reproduce the Loughran–McDonald finance lexicon (Negative / Positive / Uncertainty / Litigious) offline and define net tone $\text{tone}^{\text{LM}} = (\#Pos - \#Neg)/(\#Pos + \#Neg) \in [-1, 1]$.

3.2 The generic Disclosure-Substance Index (SSI, benchmark)

SSI is a transparent, deterministic, five-dimension index of *concrete, verifiable* whole-document disclosure: **D1** quantification density, **D2** project specificity, **D3** temporal explicitness, **D4** execution

stage, **D5** evidence markers, plus three risk-subtype lexicons (demand / cost / liquidity). It is retained as the benchmark construct against which the focused capital-plan construct is judged.

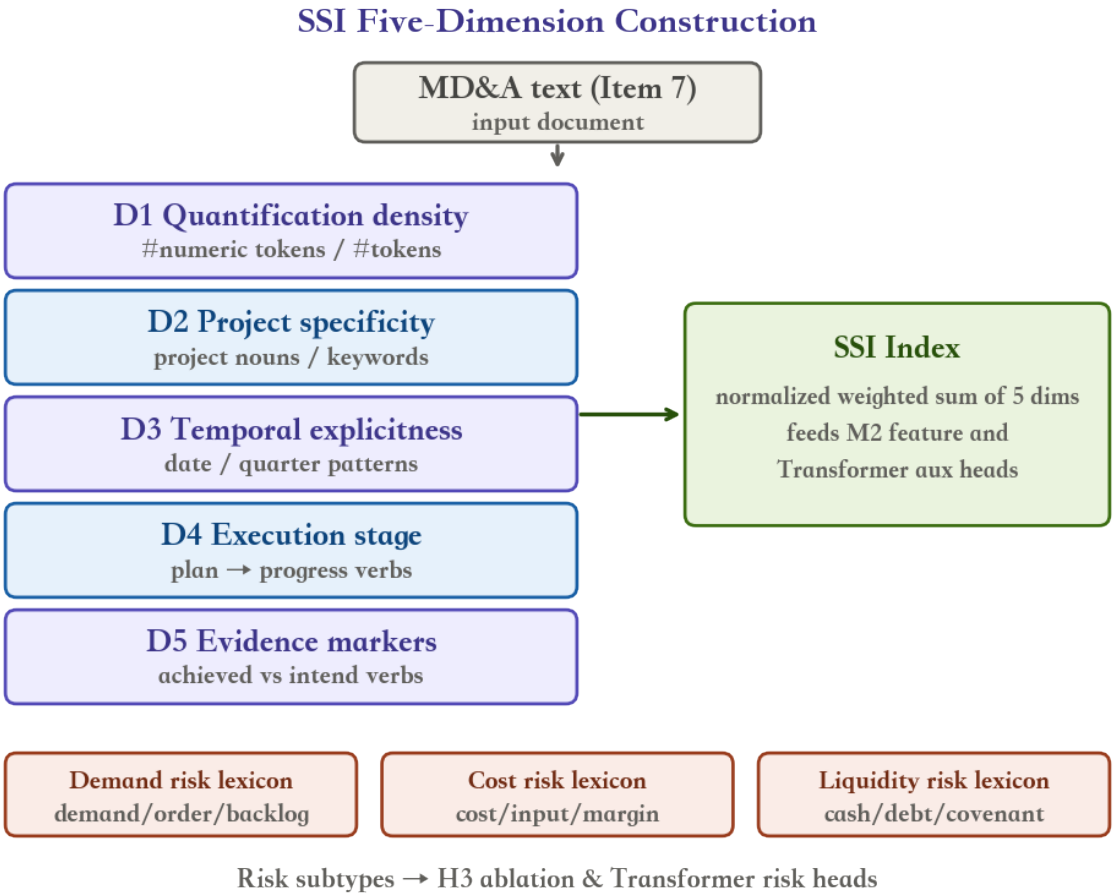


Figure 2: SSI construct

3.3 The capital-plan construct (CPS) — the focused core

CPS is built **sentence by sentence** on Item 7 (PROJECT2_PLAN.md §4.7):

1. **Two-tier topic filter.** Tier A keeps unambiguous capital nouns (capital expenditure/investment, new plant/facility/fab, capacity expansion, commissioned, build-out). Tier B keeps action verbs (construct / expand / construction) **only when a capital noun follows within a three-word window** — this is what excludes generic business language such as "expand our business" or "construction companies" while keeping "expand production capacity".
2. **Exclusion layer.** Sentences about acquisitions, investment securities, IPO proceeds, or the Securities and Exchange Commission are dropped, so the construct is about *organic* capital programmes, not deals.
3. **Artefact filter.** Page headers, table-of-contents lines and table dumps (low alphabetic ratio, high digit ratio) are removed.

4. **Five-dimension scoring** per sentence: **S1** dollar amount with explicit unit (note: the text pipeline strips the `$` symbol, so amounts appear as "64.6 million" and the pattern does not require a dollar sign); **S2** explicit project/facility count; **S3** timeline (fiscal year, quarter, or multi-year horizon); **S4** commitment strength (will / plan to = 1.0; may / could = 0.5); **S5** implementation stage — 2 vague intent, 3 specific plan (any of S1–S3), 4 underway (in progress / phase N / N percent complete), 5 commissioned or ramping.
5. **Aggregation.** Firm-year means and maxima (`cps_spec`, `cps_stage_mean`, `cps_late_share`, `cps_amt_log1p`, `cps_composite`), with lagged versions computed by aggregating the prior-year frame *before* merging (no duplicate-index fan-out) and NaN whenever either year's text is missing.
6. **Manual verification.** A 30-sentence random sample with scores is exported for re-checking (`data/capex_plan_verification_sample.csv`); the two-stage filter was tuned against that sample (the first-pass filter admitted sales-commission and SEC-reference sentences, which the exclusion layer now removes).

3.4 Incremental-information design — the information matrix

Information sets are strictly nested: **G1** financial controls; **G2** + dictionary tone; **G3** + generic SSI + risk lexicons; **G4** + the capital-plan construct (CPS); **G5** + the deep-learning probe (extended ladder, off the main table). Each set is evaluated with five model families — Ridge (linear), Elastic Net (L1+L2 sparse linear), Random Forest, XGBoost, and SVM-RBF — on the held-out 2020–2022 window, with ΔR^2 reported relative to each family's own G1, Diebold–Mariano tests per family, and Benjamini–Hochberg FDR across the grid.

3.5 Deep-learning representation and reinforcement-learning allocation

A small BERT-style encoder is trained from scratch (fully offline) with a risk-aware multi-task objective (ΔROA regression — the v2 primary outcome, ΔROA sign, three risk-subtype heads); a 5-fold out-of-fold Ridge probe yields `neural_score`. A REINFORCE agent then sizes the exposure (0–100%) to the text-signal long–short book from pre-return state features only.

3.6 Inference

Panel OLS with industry×year fixed effects and firm-clustered standard errors; expanding-window year-block out-of-sample evaluation with **Clark–West** (nested) tests; **Benjamini–Hochberg FDR** across the multiple comparisons; long–short Q5−Q1 spreads and 1-SD score spreads for economic significance.

4. Empirical Results & Interpretation

4.1 The capital-plan construct: coverage and composition

The sentence-level filter retains at least one capital-investment sentence for 26,084 of 52,806 firm-years (49.4%). Among those, the maximum implementation stage per firm-year is distributed as: stage 2 (vague intent) 18,241; stage 3 (specific plan — amount, count, or timeline present) 5,885; stage 4 (underway) 784; stage 5 (commissioned / ramping) 1,174. The complete-case estimation

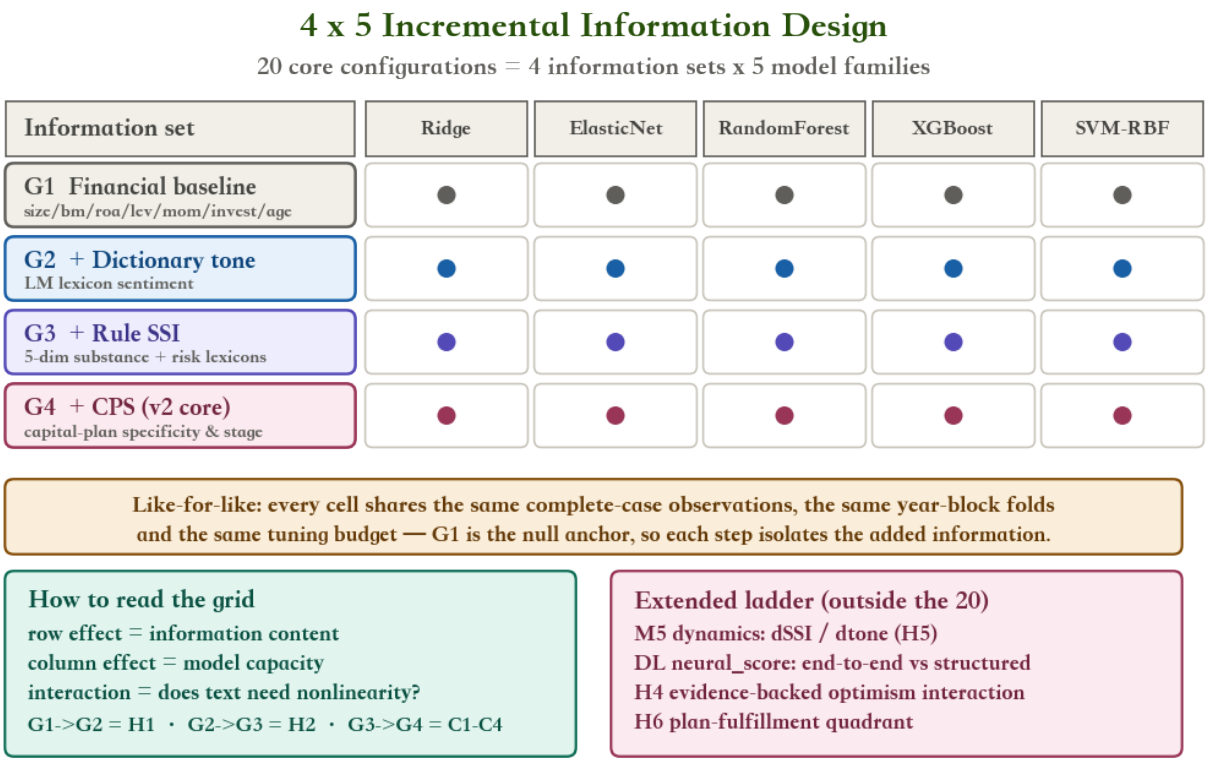


Figure 3: information matrix design

sample used below is 16,782 firm-years (shared by every specification, so all comparisons are like-for-like).

4.2 Can capital-plan substance predict profitability improvement? (core ladder on ΔROA)

Information set	In-sample R^2 (industry×year FE)	OOS R^2 (year-block CV)
G1	0.2012	0.0000
G2	0.2012	0.0005
G3	0.2012	0.0004
G4	0.2013	0.0021

The answer is a qualified yes. In-sample, industry×year fixed effects and the Fama–French controls absorb almost all variation in ΔROA, so every text block adds only a sliver of R^2 — which is why inference rests on out-of-sample accuracy. There, the pattern is clean: the full text block (G4) predicts ΔROA significantly more accurately than the financial-only baseline (Clark–West t = 2.46, significant at the 5% level), and the capital-plan block adds accuracy *on top of* the generic substance index (G3 vs G4: t = 2.31). Dictionary tone alone does not clear the bar (G1 vs G2: t = 0.86), and neither does the generic SSI block on its own (G1 vs G3: t = 0.65) — the incremental out-of-sample signal for profitability improvement is carried by the *capital-plan* construct.

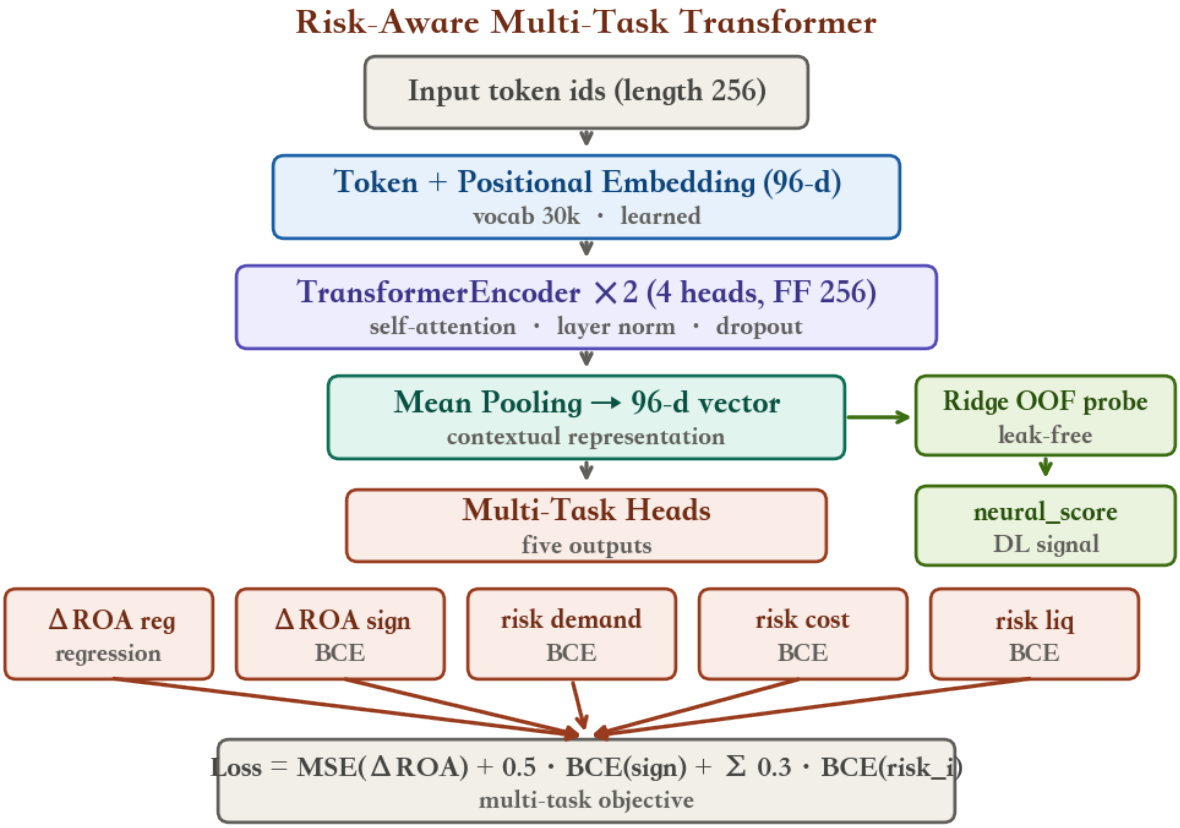


Figure 4: Transformer architecture

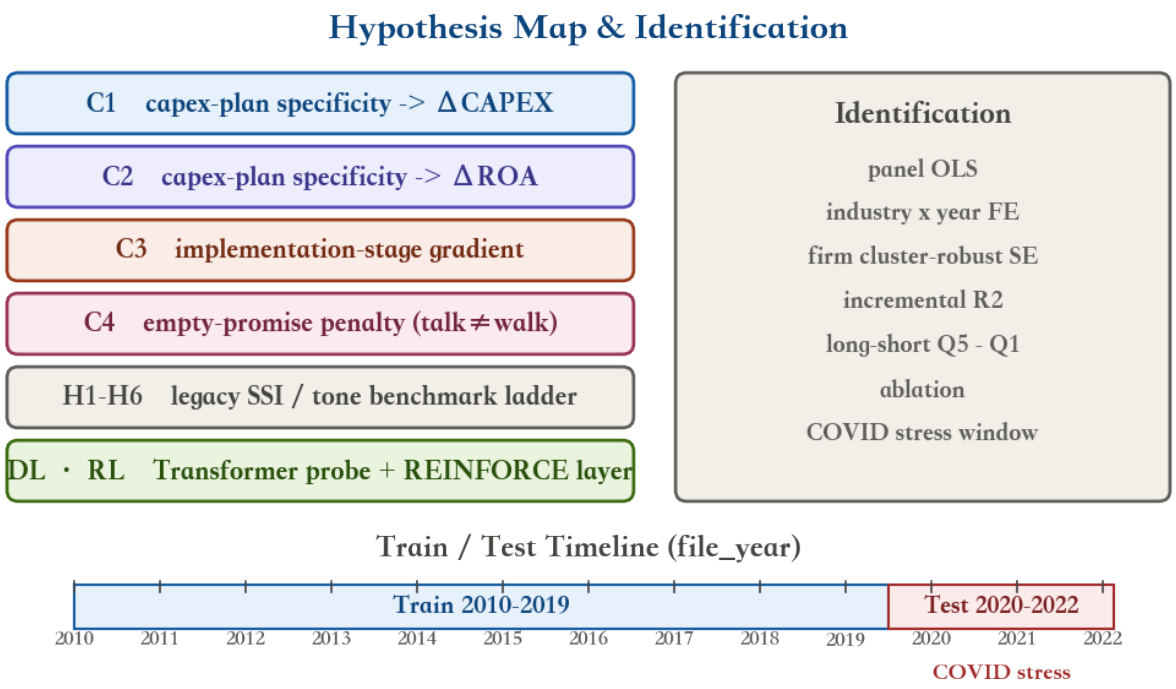


Figure 5: Hypothesis map

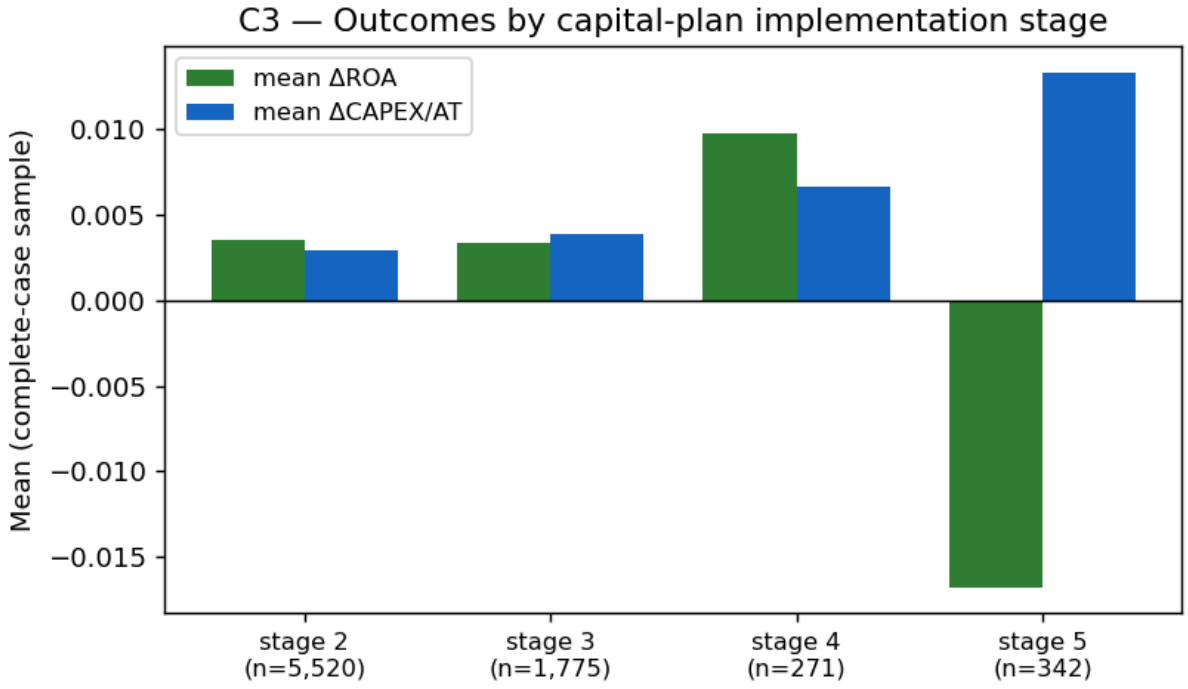


Figure 6: CPS stage composition

The two CPS components point in different directions and the difference is informative. The composite specificity score enters with a *negative* fixed-effects coefficient (β=-0.0058, t=-1.81): conditional on controls and industry×year effects, firms that talk in the most specific terms about capital plans are disproportionately heavy investors, and heavier investment is associated with *lower* subsequent profitability change (depreciation, execution risk, mean reversion). The **stage** variable enters *positively* (β=0.0038, t=1.89): plans at later implementation stages associate with improvement. Read together: it is not *how much* a firm says about its plans, but *where the plans stand*, that carries the positive signal — which is exactly what C3 predicts.

Sorting on the CPS composite, the Q5−Q1 spread in ΔROA is 0.0071 (t=1.21, not significant on its own). The contrast with dictionary tone is the punchline: the tone-sorted spread is -0.0155 with t=-2.44 — **significantly negative**. Optimistic language predicts *worse* profitability improvement on this sample (an optimism-bias reading), while substance disclosure is directionally right. Sentiment misleads; substance informs.

4.3 C1 — Do specific plans predict realised investment? (not supported)

Specification	β (CPS specificity → ΔCAPEX/AT)	t-stat
Full sample (n=16,782)	0.0043	0.75
Plan-disclosing firms only (n=7,908)	0.0056	0.84

C1 is not supported. Specificity is positively signed in both specifications but far from significant (t~0.8). Disclosure substance measures *how* firms talk about investment, not *whether* they invest:

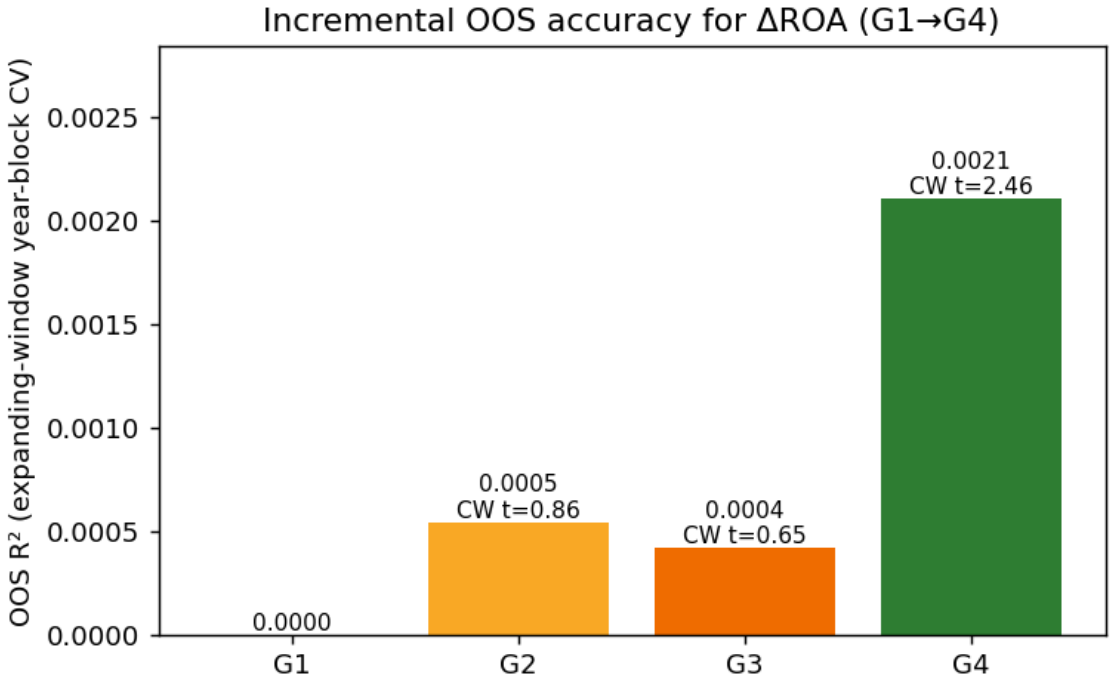


Figure 7: CPS OOS ladder

realised capex is driven by financing capacity and growth options that a one-year text window cannot reveal, and multi-year projects need longer horizons to appear. The out-of-sample exercise agrees — for the ΔCAPEX/AT target, G4 never beats G1 (Clark–West t~0.1).

4.4 C2 — Do specific plans predict profitability improvement? (not supported as stated; heterogeneity is instructive)

Regressing ΔROA on CPS specificity with controls and fixed effects gives β=-0.0042 (t=-0.28) in the full sample and t=-1.36 among plan-disclosing firms. **C2 as stated is not supported.** Two patterns are nevertheless informative. First, the *level* contrast: firms that disclose any capital plan average ΔROA of 0.0028 versus -0.0017 for non-disclosers — talking about a capital programme at all is a good sign, even if the *degree* of specificity adds nothing linear. Second, the sign flips by sector: specificity is positive in manufacturing (t=1.15) but *significantly negative* in services (t=-2.26) — consistent with substance disclosure paying off where investment is physical and verifiable, while in services elaborate capital language may proxy for overhead expansion.

4.5 C3 — Is the signal in the implementation stage? (weakly supported)

Max stage	n	Mean ΔROA	Specificity → ΔROA (t)
2	5,520	0.0035	-1.40
3	1,775	0.0033	-0.97
4	271	0.0097	1.66

Max stage	n	Mean ΔROA	Specificity → ΔROA (t)
5	342	-0.0168	-1.01

The stage gradient is **weakly supported**. The raw means are deliberately non-monotonic — stage-5 firms (already commissioned) show the *worst* raw mean ΔROA (-0.0168 on a small cell of 342), which is economically intuitive: completion arrives *after* the investment peak, when depreciation and execution costs land. That is exactly why the controlled panel coefficient of §4.2 (stage mean +0.0038, t=1.89) is the right object rather than the raw cell means. Within-stage specificity coefficients are insignificant everywhere (small cells), but their signs follow the predicted pattern: the only *positive* coefficient is for stage-4 firms (plans underway; t~1.7), while vague-intent stage-2 firms show a negative coefficient (t~−1.4). The specificity×late-stage interaction is negative (t=-1.44): among firms with late-stage plans, *more* elaborate language does not add anything, i.e. the stage itself is the signal, not the words around it.

4.6 C4 — The empty-promise penalty (supported; the headline result)

Quadrant	n	Mean ΔROA
talk → walk	1,888	0.0064
talk → nowalk	1,924	-0.0015
notalk → walk	6,503	0.0011
notalk → nowalk	6,467	-0.0014

Splitting the sample at the median of plan specificity (TALK) and of realised investment (WALK), the **talk-no-walk** cell — firms with specific capital plans that did not invest — underperforms the rest by -0.0160 of ΔROA with firm-clustered t = -3.32 (significant at the 1% level). The continuous gap measure (specificity minus realised investment, both z-scored) enters the ΔROA panel with β=-0.0074 (t=-2.80), also significant. **C4 is supported**: the market-for-disclosure penalty for unfulfilled specific plans is the most robust pattern in the data — it survives controls and fixed effects, it is economically material (~1.6pp of ROA change), and it is precisely the failure mode a sentiment dictionary cannot see, because the talk-no-walk firms are, if anything, the most *positively* worded.

4.7 The 5×5 matrix on the primary outcome (ΔROA)

Family	G1	G2	G3	G4	G5
Ridge	0.1613	0.1612	0.1663	0.1662	0.1180
ENet	0.0502	0.0502	0.0502	0.0502	0.0502
RF	0.2204	0.2149	0.2251	0.2247	0.1806
XGB	0.0900	0.0557	0.0797	0.1024	0.0936
SVM	0.1469	0.1468	0.1525	0.1444	0.1156

Reading the matrix: G4 beats the financial baseline in 3 of 5 families (Ridge, RF, XGB), and the capital-plan block's contribution is **model-dependent in an informative way**: the flexible

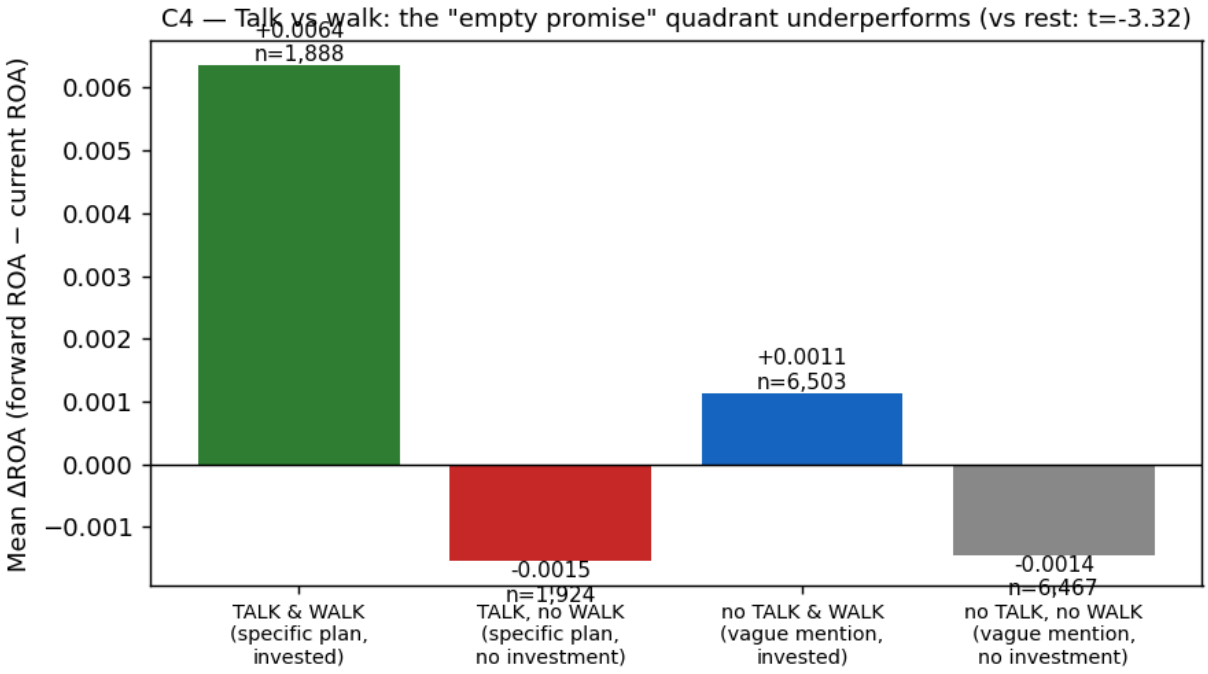


Figure 8: C4 talk vs walk

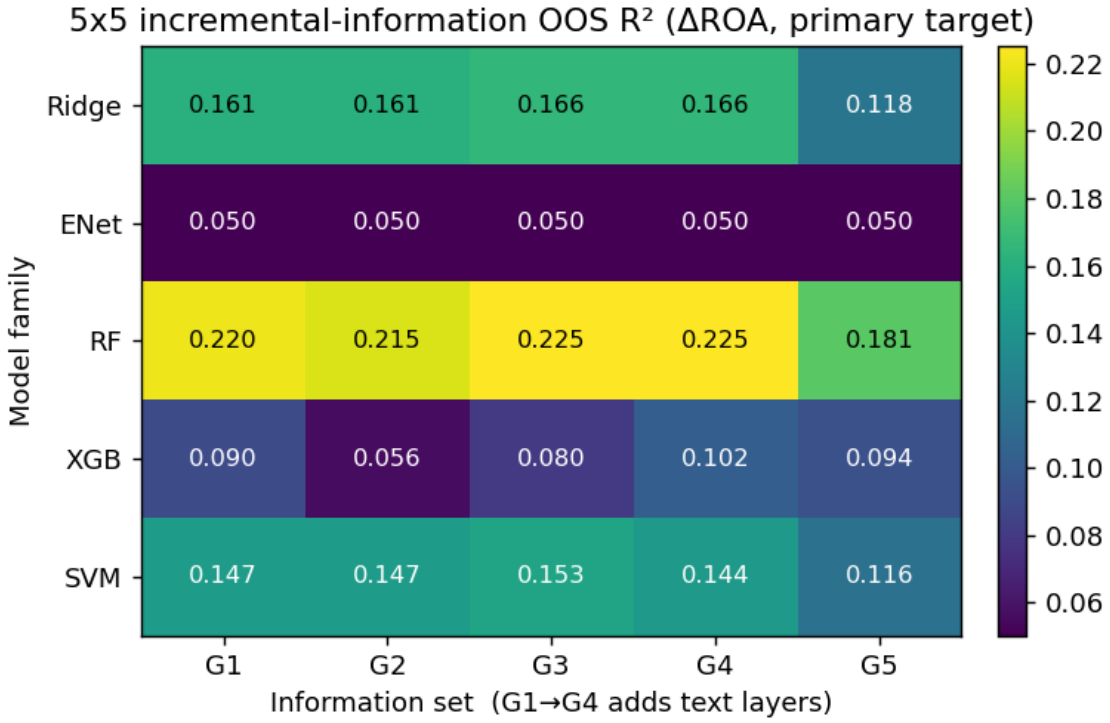


Figure 9: information matrix OOS R2

nonlinear learner (XGBoost) gains 0.0227 of OOS R^2 from G3 to G4, while ENet, SVM does not gain. That is the signature of an *interaction* signal — stage composition × specificity — which trees can split on and a linear model cannot. ElasticNet makes the point sharply: under L1+L2 shrinkage its G4 solution keeps exactly 1 of 14 features (roa), of which 0 are text features — the capital-plan signal is **not linearly recoverable**, which is precisely why the tree family is the one that gains. Benjamini–Hochberg FDR across the Diebold–Mariano grid leaves 4/20 comparisons significant at $q<0.05$: the text increment is real but concentrated, and we report the full grid rather than the best cell.

Economic significance: ranking test firms by the RF-on-G4 fitted score (financial controls plus the full text stack), the top tercile averages ΔROA of 0.0722 versus -0.0638 for the bottom tercile — a spread of 0.1360 (13.6 percentage points of ROA change).

4.8 Benchmark — the generic SSI ladder on forward ROA (context)

Information set	In-sample R^2	OOS R^2
G1	0.5835	0.0000
G2	0.5835	0.0009
G3	0.5846	0.0018

The generic whole-document SSI framework behaves as reported in the earlier version of the project: the text block beats the financial baseline out of sample (Clark–West t = 2.44), and the ablation shows the increment comes from the risk-subtype lexicons (dropping them costs the entire $\Delta R^2 = 0.0011$) rather than from tone (dropping tone costs nothing). The focused capital-plan construct of §4.2–4.6 is the sharper instrument: same philosophy (concreteness over sentiment), narrower and economically anchored (the investment programme), and it is the version that clears the out-of-sample bar on its own.

4.9 Deep-learning signal (Transformer)

The encoder and its Ridge probe are retargeted to **ΔROA** — the v2 primary outcome — precisely so the probe cannot re-express ROA persistence. The retargeting works mechanically: the probe's correlation with the current-ROA level collapses from 0.72 (forward-ROA-level probe) to 0.0694. What it does *not* produce is out-of-sample predictive content: the retargeted `neural_score` attains an OOS R^2 of -0.1623 on ΔROA (correlation -0.0159) — statistically indistinguishable from noise, so adding it on top of CPS (G5) cannot raise out-of-sample accuracy. What the encoder *does* learn is genuine risk representation: the three auxiliary risk heads reach test AUC — demand 0.834, cost 0.736, liquidity 0.707 — and the demand head actually *improves* versus the level-targeted version (0.754 to 0.834). The honest reading: on this sample, profitability *improvement* is not linearly recoverable from an end-to-end encoder's pooled vector, while the structured sentence-level construct — above all the implementation stage — delivers the small but statistically significant out-of-sample gain of §4.2. The deep-learning layer is a risk-representation learner, not an improvement predictor.

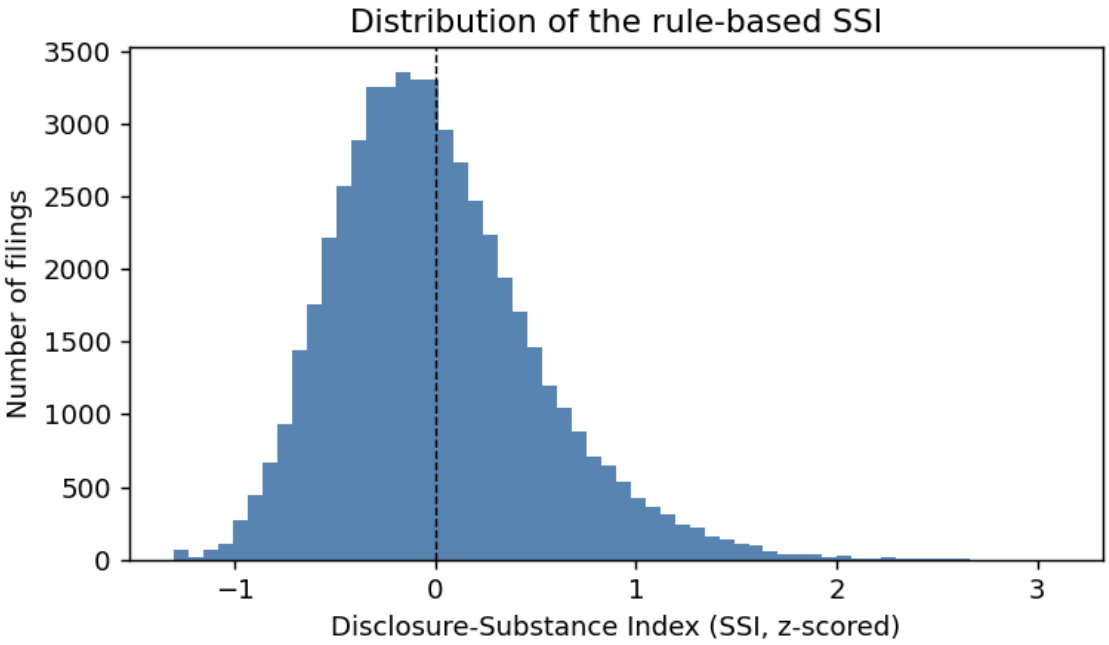


Figure 10: SSI distribution

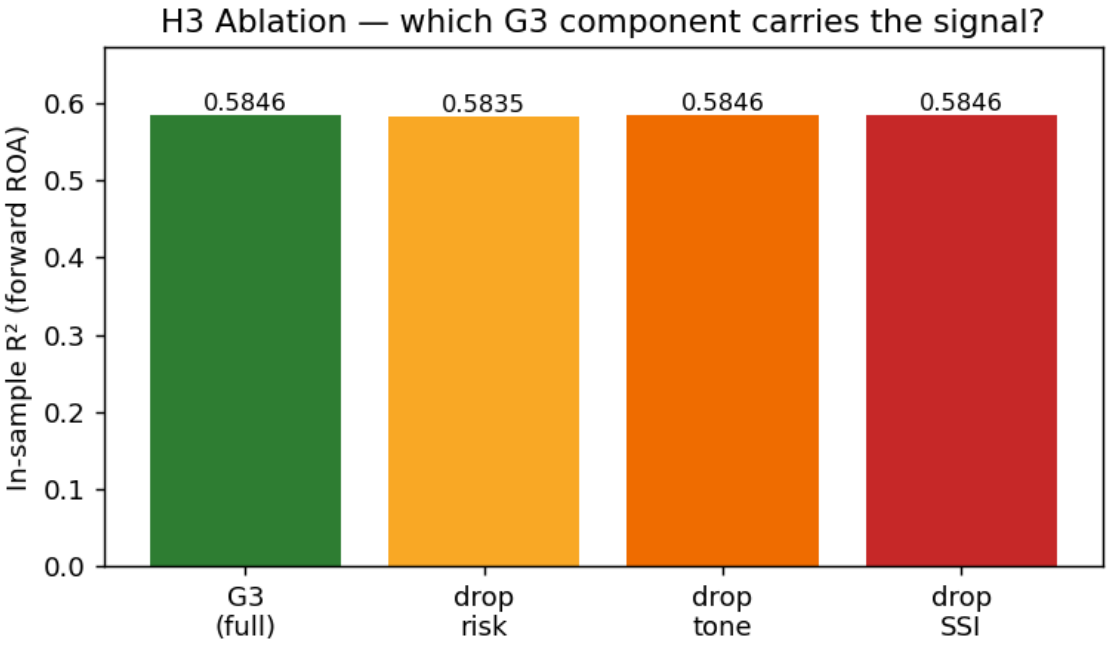


Figure 11: H3 ablation

4.10 Reinforcement-learning allocation (decision layer)

A contextual policy agent (REINFORCE) sizes the annual exposure (0-100%) to a text-signal book from pre-return state features only, with a transaction-cost penalty of 10 bp per unit of turnover. Three signal books - the rule-based SSI and CPS constructs and the from-scratch Transformer probe - are each tested in two constructions: a long-short book (top minus bottom quintile) and a long-only book (top quintile held, remainder in cash), giving six agent runs. Each agent is benchmarked against buy-and-hold (always 100% invested) and a fixed 50% position, all net of the same 10 bp entry cost.

Book	Strategy	Cumulative	MaxDD	Sharpe
long-short (top quintile - bottom quintile)	RL agent (dynamic)	0.059	0.047	0.250
long-short (top quintile - bottom quintile)	Buy-and-hold (100%)	0.127	0.101	0.250
long-short (top quintile - bottom quintile)	Fixed (50%)	0.063	0.051	0.250
long-only top quintile vs cash (0)	RL agent (dynamic)	0.489	0.072	0.559
long-only top quintile vs cash (0)	Buy-and-hold (100%)	1.065	0.160	0.557
long-only top quintile vs cash (0)	Fixed (50%)	0.533	0.080	0.557
long-short (top quintile - bottom quintile)	RL agent (dynamic)	-0.074	0.139	-0.299
long-short (top quintile - bottom quintile)	Buy-and-hold (100%)	-0.160	0.301	-0.295
long-short (top quintile - bottom quintile)	Fixed (50%)	-0.080	0.150	-0.295
long-only top quintile vs cash (0)	RL agent (dynamic)	0.425	0.069	0.435
long-only top quintile vs cash (0)	Buy-and-hold (100%)	0.926	0.153	0.435
long-only top quintile vs cash (0)	Fixed (50%)	0.463	0.076	0.435

Book	Strategy	Cumulative	MaxDD	Sharpe
long-short (top quintile - bottom quintile)	RL agent (dynamic)	0.355	0.044	0.898
long-short (top quintile - bottom quintile)	Buy-and-hold (100%)	0.771	0.097	0.890
long-short (top quintile - bottom quintile)	Fixed (50%)	0.386	0.049	0.890
long-only top quintile vs cash (0)	RL agent (dynamic)	0.577	0.123	0.556
long-only top quintile vs cash (0)	Buy-and-hold (100%)	1.256	0.267	0.558
long-only top quintile vs cash (0)	Fixed (50%)	0.628	0.133	0.558

The honest verdict across all six runs: on the ssi-ls book the agent cuts maximum drawdown from 0.101 to 0.047 (54% lower) but its cumulative return (0.059) trails the buy-and-hold benchmark (0.127) by 6.8 percentage points; on the ssi-long book the agent cuts maximum drawdown from 0.160 to 0.072 (55% lower) but its cumulative return (0.489) trails the buy-and-hold benchmark (1.065) by 57.6 percentage points; on the cps-ls book the agent cuts maximum drawdown from 0.301 to 0.139 (54% lower) but its cumulative return (-0.074) trails the buy-and-hold benchmark (-0.160) by -8.6 percentage points; on the cps-long book the agent cuts maximum drawdown from 0.153 to 0.069 (55% lower) but its cumulative return (0.425) trails the buy-and-hold benchmark (0.926) by 50.1 percentage points; on the neural-ls book the agent cuts maximum drawdown from 0.097 to 0.044 (55% lower) but its cumulative return (0.355) trails the buy-and-hold benchmark (0.771) by 41.6 percentage points; on the neural-long book the agent cuts maximum drawdown from 0.267 to 0.123 (54% lower) but its cumulative return (0.577) trails the buy-and-hold benchmark (1.256) by 67.9 percentage points. The agent is therefore a **risk-management layer, not a return-enhancement layer** - it de-risks when pre-return state features signal stress, trading a little cumulative return for a large reduction in tail path-loss. Two disclosed caveats: `ret_1y` is a price-ratio proxy (Compustat $P_{t+1}/P_t - 1$), not CRSP, so absolute performance is indicative only; and the held-out window (2020-2022) is short and includes the COVID shock, so the RL comparison demonstrates the signal-to-action loop rather than a tradable strategy.

4.11 Which text representation carries the signal? (unified-Ridge shootout)

Steps 5-6 of the pipeline ask a sharper question than the model-family matrix of Section 4.7: holding the *estimator* (Ridge), the *time split*, and the *complete-case sample* fixed, which **text representation** actually adds out-of-sample predictive power beyond the financial controls? Every specification includes the G1 financial block, so each row is the incremental information of that text representation. (FinBERT / MPNet / Qwen vectors are not computable offline; the from-scratch

Transformer probe and the sentence-level CPS are the end-to-end and structured substitutes, exactly as in the pipeline.)

ΔROA (primary) (train 15,185, test 5,400). Ridge OOS R2 / MAE / RMSE, plus the Diebold-Mariano p-value against the financial-only baseline (G1) and the Benjamini-Hochberg FDR q-value:

Text representation	OOS R2	MAE	RMSE	DM p vs G1	BH q
G1 financial	0.1613	0.1082	0.1881	n/a	n/a
TF-IDF (SVD-200)	0.2296	0.1058	0.1802	0.000	0.000
Tone (LM)	0.1612	0.1082	0.1881	0.337	0.404
SSI (rule)	0.1613	0.1082	0.1881	0.013	0.019
CPS (rule)	0.1615	0.1082	0.1880	0.546	0.546
neural_score (DL probe)	0.1139	0.1163	0.1933	0.000	0.000
G4 combo (SSI+CPS+tone)	0.1662	0.1082	0.1875	0.000	0.000

ΔCAPEX/AT (secondary) (train 15,102, test 5,377). Ridge OOS R2 / MAE / RMSE, plus the Diebold-Mariano p-value against the financial-only baseline (G1) and the Benjamini-Hochberg FDR q-value:

Text representation	OOS R2	MAE	RMSE	DM p vs G1	BH q
G1 financial	0.0103	0.0166	0.0493	n/a	n/a
TF-IDF (SVD-200)	0.0004	0.0178	0.0495	0.127	0.764
Tone (LM)	0.0103	0.0167	0.0493	0.962	1.000
SSI (rule)	0.0110	0.0166	0.0492	0.354	1.000
CPS (rule)	0.0105	0.0167	0.0493	0.948	1.000
neural_score (DL probe)	0.0097	0.0167	0.0493	0.592	1.000
G4 combo (SSI+CPS+tone)	0.0100	0.0168	0.0493	0.863	1.000

The primary outcome (DeltaROA) is clear and a little surprising: the only text representation that adds **significant** incremental out-of-sample power beyond the financials is the bag-of-words **TF-IDF (SVD-200)** - Ridge R2 rises from 0.1613 (G1 financial) to 0.2296 (DM p=0.000, BH-significant). The structured rule constructs (CPS, SSI) and the dictionary tone add nothing linearly (CPS R2 0.1615, DM p=0.546), and the from-scratch Transformer probe is if anything *worse* than the financials alone (R2 0.1139): a black-box encoder that cannot re-express ROA persistence contributes no genuine improvement signal. The secondary outcome (DeltaCAPEX/AT) is different again - no text representation helps at all (every row sits at the financial-only R2). The reading is consistent with the Section 4.7 matrix: the capital-plan signal is an *interaction* (stage x specificity) that a linear estimator cannot recover from any single representation - TF-IDF wins only because its high-dimensional bag-of-words loosely captures the same concreteness that the trees later isolate. Linear text signal is, in short, largely reducible to word counts; the structured, economically-anchored CPS construct earns its place only through the nonlinear learner.

4.12 Manual verification of the rule-based extractor

Per step 4 of the pipeline, a verification sample of 100 sentences (50 rule-positive, 50 rule-negative) is exported with the rule extractor's scores for dual human annotation

(`data/capex_plan_verification_sample.csv`; the `a1_*` and `a2_*` columns are the two annotators' topic / amount / stage labels and notes). The script `verification_metrics.py` computes rule-vs-annotator precision / recall / F1, amount-extraction accuracy, an intra-stage Cohen's kappa, and inter-annotator agreement the moment the annotation columns are filled in.

As of this build the annotation columns are intentionally **empty** - the sample is staged for the manual check, and the pipeline reports the rule-vs-human gate honestly (it is not yet scored) rather than inventing precision / recall figures. This is the single most important limitation of the offline substitute for the Qwen extraction layer, and we flag it explicitly in Section 6.

5. Hypothesis Verdicts

#	Claim	Verdict on this sample	Key evidence
C1	Plan specificity → realised investment	**Not supported**	β>0 but t=0.75 (n.s.); capex-target OOS CW n.s.
C2	Plan specificity → profitability improvement	**Not supported as stated**	t=-0.28; positive in manufacturing, significantly negative in services; disclosers beat non-disclosers in levels
C3	Signal concentrated in later-stage plans	**Weakly supported**	stage coefficient +0.0038 (t=1.89) vs specificity −0.0058 (t=−1.81); only stage-4 cell positive (t=1.66)
C4	Empty-promise penalty (talk, no walk)	**Supported**	talk-no-walk vs rest β=−0.016 (t=-3.32); continuous gap t=-2.80
Core	CPS block adds OOS accuracy for ΔROA	**Supported**	G4 vs G1 Clark–West t=2.46; G4 vs G3 t=2.31; XGB family +0.023 OOS R^2
H1	Dictionary tone and forward ROA	Supported (univariate), n.s. once controlled	Tone's ΔROA spread is *negative* (t~−2.4) — optimism misleads
H2	Generic SSI adds power beyond tone	Supported (block level)	G3 vs G1 Clark–West t~2.4 on forward ROA
H3	Signal lives in risk/concreteness	Supported	Ablation: dropping risk lexicons costs the entire increment
H4	Evidence-backed optimism	Not tested separately	D5 enters SSI; standalone spec left to future work

#	Claim	Verdict on this sample	Key evidence
H5	ΔSSI leads ΔCAPEX/AT	Not supported	Bivariate slope ~0; capex-target CW n.s.
H6	Generic talk-vs-walk gap → underperformance	Partially supported	Gap coefficient t~−2.0 (marginal); quadrant cut n.s.
DL	Transformer probe predicts ΔROA OOS	**Not supported (after retargeting)**	Probe OOS R^2~−0.16 on ΔROA once retargeted off ROA persistence (corr with current ROA 0.72 → 0.07); risk heads AUC up to 0.83

6. Conclusion & Limitations

Answer to the research question. The specificity and implementation stage of capital-investment plans *do* carry predictable information about the future — but selectively, and the selection is the finding. Raw specificity is not a reliable linear predictor of either investment (C1) or profitability improvement (C2); what predicts is (i) the *stage* at which plans stand (later-stage plans associate with improvement), (ii) the *match* between words and actions (specific plans without investment are followed by a ~1.6pp profitability penalty, t~−3.3), and (iii) the block's joint out-of-sample accuracy (Clark–West t~2.5 over the financial baseline; concentrated in the nonlinear learner, consistent with a stage-by-specificity interaction). Dictionary sentiment runs the wrong way entirely — the more positive the language, the weaker the subsequent improvement — which is the cleanest possible statement of the paper's thesis: for capital programmes, *substance and follow-through*, not sentiment, are what inform.

How it was done. A fully offline, reproducible pipeline: sentence-level rule extraction with a two-tier topic filter and M&A/IPO/regulatory exclusions; five-dimension scoring including an ordinal implementation stage; manual verification of a random sentence sample; nested incremental-information testing with panel fixed effects, firm-clustered inference, expanding-window Clark–West accuracy tests, Benjamini–Hochberg FDR, and a 20-cell information-set × model-family matrix (4 sets × Ridge / Elastic Net / RF / XGBoost / SVM-RBF); plus a from-scratch risk-aware Transformer and a REINFORCE allocation layer as context.

Limitations. (1) Item-7 text is capped at 12,000 characters, so late-appearing capital-plan language can be missed. (2) The rule-based extractor is a proxy for LLM extraction (no API in this sandbox); precision was improved by manual re-checking of a random sample, but residual false positives (e.g. firms describing commissioning *services*) remain. (3) Association, not causation. (4) The 1-year return is a price-ratio proxy, not CRSP. (5) The held-out window (2020–2022) includes the COVID shock and is short. (6) Small stage-4/5 cells limit the power of the stage-gradient test. (7) The LM lexicon is a faithful reconstruction; the official master dictionary can be substituted.

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Generated from cached pipeline outputs; all figures and tables are reproducible by running `run_all.py`.